

Trainings ▾

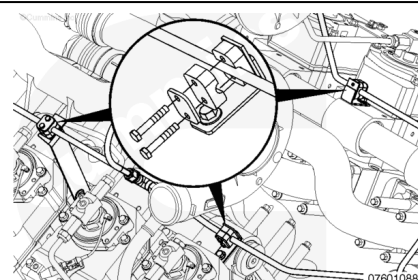
Remove

QSK45 and QSK60 Single Stage

Note : The steps listed below apply to both banks of the engine.

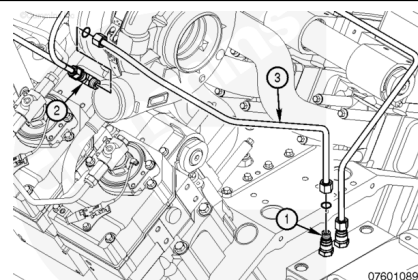
With Mechanically Actuated Injector

- Remove the two capscrews and clamps from the oil supply lines.



⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.



⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be a carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ CAUTION ⚠

Observe the routing, clipping and clearances of fluid plumbing lines and presence of any shielding in order to reference when reinstalling.

Disconnect the oil supply tube from the reducer (1) on the flywheel housing. Discard the o-ring.

Disconnect the oil supply tube from the T-fitting (2). Discard the o-ring.

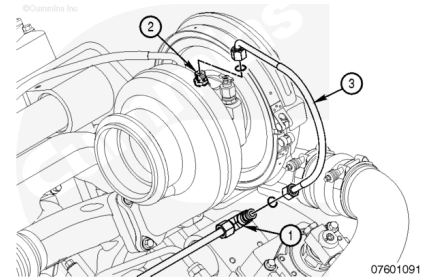
Remove the oil supply tube (3).

Disconnect the turbocharger oil supply tube from the T-fitting (1). Discard the o-ring.

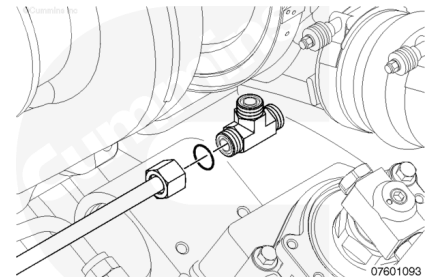
Disconnect the oil supply tube from the male union (2).

Discard the o-ring.

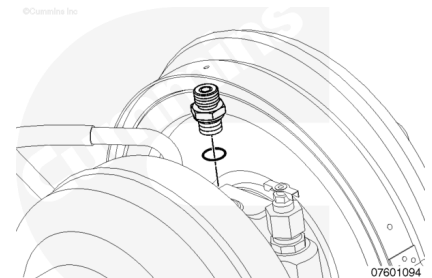
Remove the oil supply line (3).



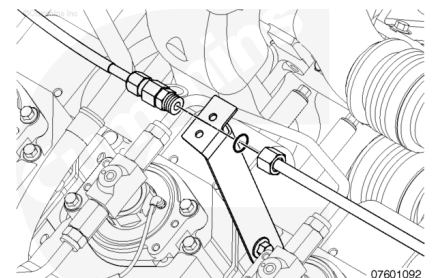
Disconnect the oil supply hose from the T-fitting and discard the o-ring.



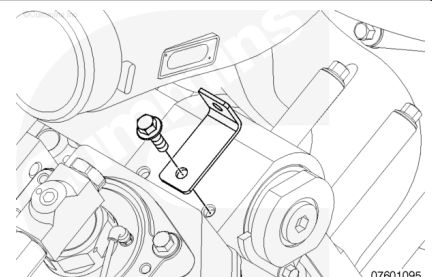
Remove the union from the top of the turbocharger and discard the o-ring.



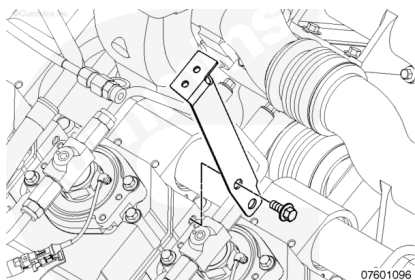
Disconnect the oil supply hose from the reducer and union. Discard the o-ring.



Remove the capscrew securing the clamp to the engine block.



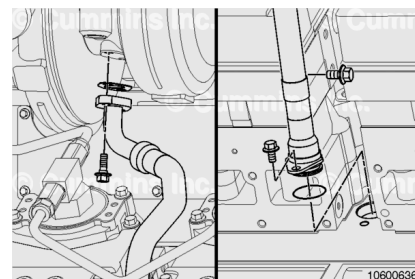
Remove the two capscrews securing the braces to the engine block. Remove the brace.



With Electronically Actuated Injector

- Disconnect the clamps securing the turbocharger oil drain tube.

Note : The steps listed below apply to both banks of the engine.

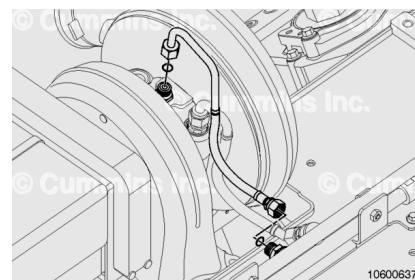


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⚠ WARNING ⚠

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⚠ CAUTION ⚠

Observe the routing, clipping and clearances of fluid plumbing lines and presence of any shielding in order to reference when reinstalling.

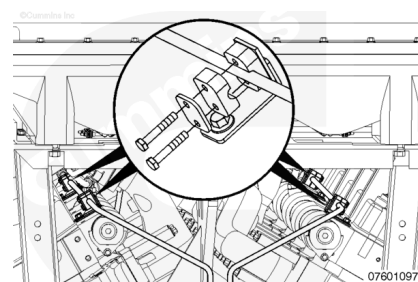
Disconnect the oil supply tube from the turbocharger and turbocharger oil manifold.

Discard the o-rings.

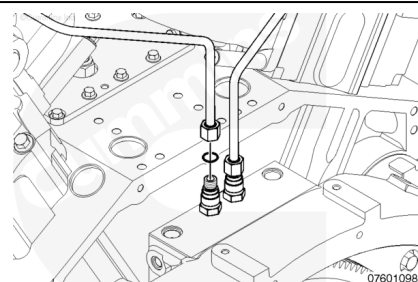
QSK60 Two Stage

Note : The steps listed below applies to both banks of the engine.

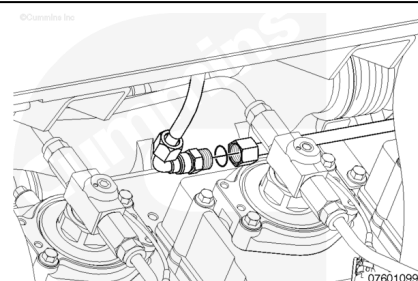
Remove the two capscrews and clamps from the oil supply lines.



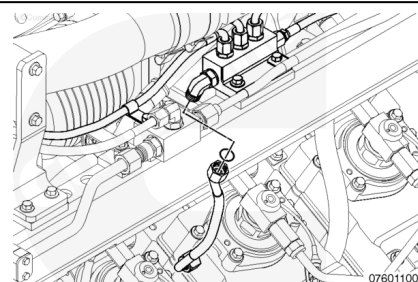
Disconnect the oil supply tube from the reducer on the flywheel housing. Discard the o-ring.



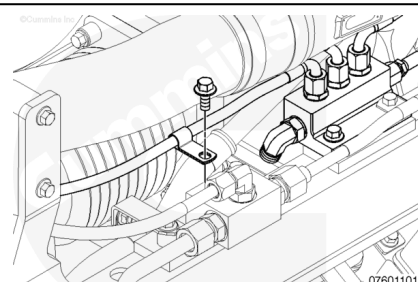
Disconnect the oil supply tube from the hose. Remove the oil supply line and discard the o-ring.



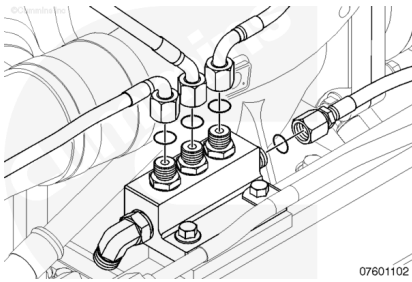
Disconnect the oil supply hose from the union on the oil supply manifold. Discard the o-ring.



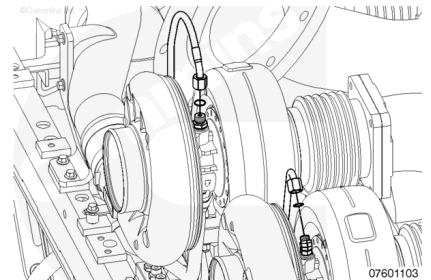
Disconnect the p-clip securing the oil supply lines from the manifolds to the turbochargers.



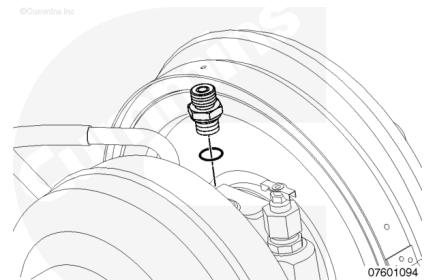
Disconnect the oil supply lines from the manifold and discard the o-rings.



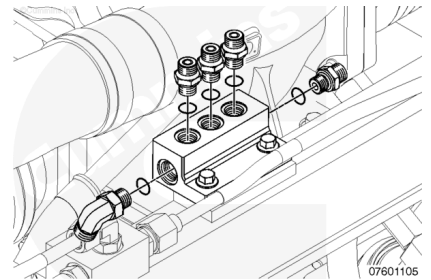
Remove the oil supply lines from the turbocharger unions and discard the o-rings.



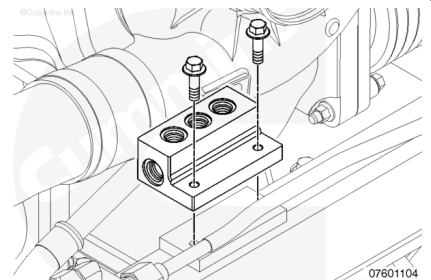
Remove the male unions from the turbochargers and discard the o-rings.



Remove the elbow and unions from the oil supply manifold. Discard the o-rings.



Remove the oil supply manifold by removing the two capscrews.



QSK45 and QSK60 Marine Single Stage With Flexible Lines

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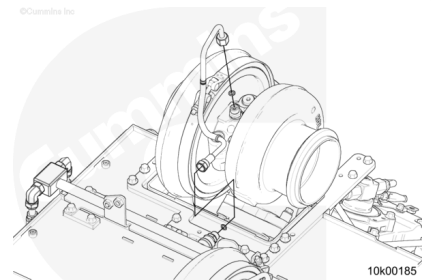
⚠ CAUTION ⚠

Observe the routing, clipping and clearances of fluid plumbing lines and presence of any shielding in order to reference when reinstalling.

Left bank rear and right bank front. Disconnect the oil supply tube from the hose.

Make sure top connection is disconnected followed by bottom (to make sure oil drains into the engine).

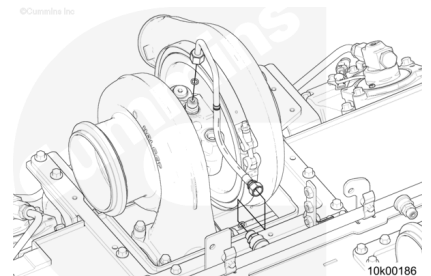
Remove the oil supply line and discard the o-ring.



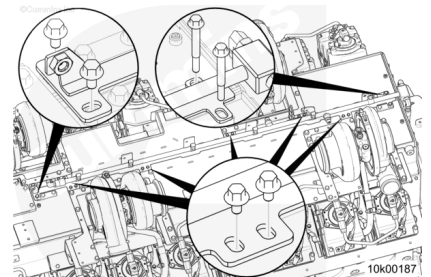
Left bank front and right bank rear. Disconnect the oil supply tube from the hose.

Make sure top connection is disconnected followed by bottom (to make sure oil drains into the engine).

Remove the oil supply line and discard the o-ring.

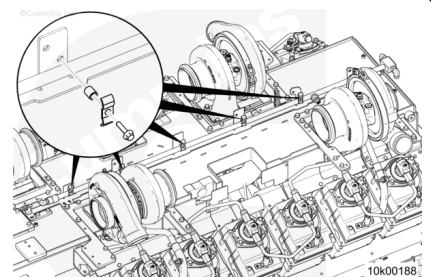


Remove capscrews.

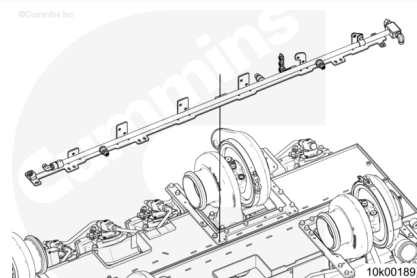


Remove five capscrews, clamps, and spacers from the transfer tube bracket.

Remove the coolant transfer tubes.

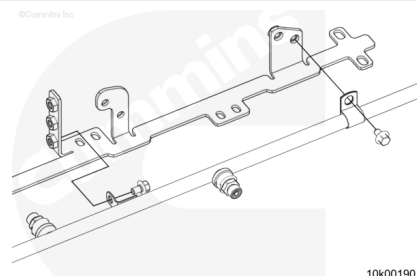


Remove the oil transfer tube and bracket assembly from the engine.

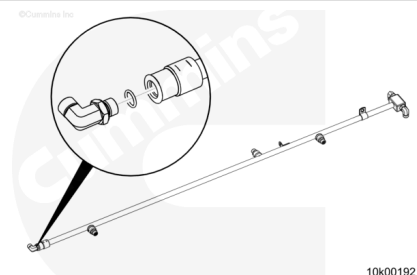


Remove the P-clip capscrew and L-bracket capscrews from the transfer tube bracket.

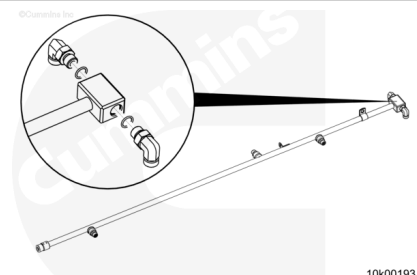
Remove the oil transfer tube from the bracket.



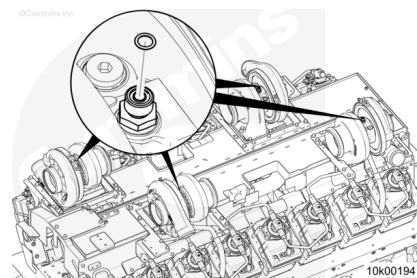
Remove elbow male union off the front port of the oil transfer tube.



Remove two elbow male unions to the rear ports of the oil transfer tube.



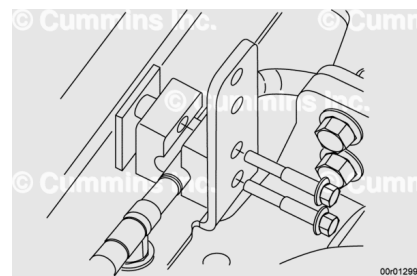
Remove and discard the o-rings on the male unions.



QSK45 and QSK60 Marine Single Stage With Rigid Lines

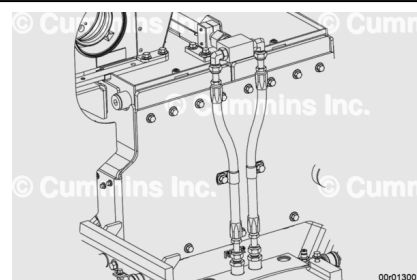
⚠ CAUTION ⚠

Observe the routing, clipping and clearances of fluid plumbing lines and presence of any shielding in order to reference when reinstalling.

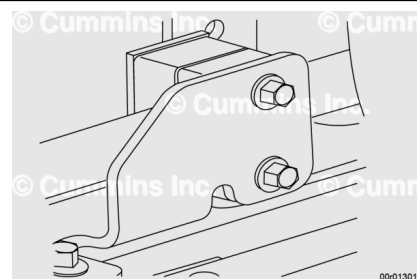


Remove connector blocks and lubricating oil supply hoses to turbocharger.

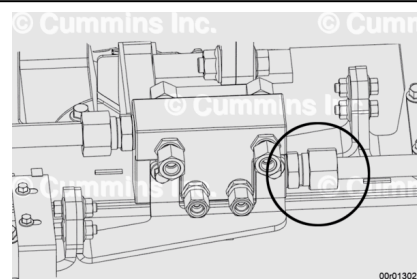
Remove both lubricating oil supply hoses from the two elbows at the rear of the engine.



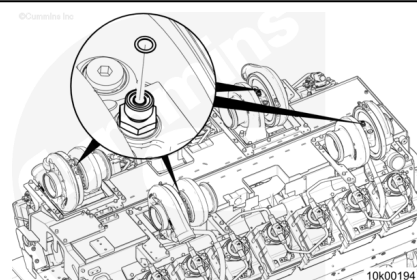
Remove the four capscrews out of the rail tube clamps supporting the lubricating oil supply tube.



Loosen the nut connecting the supply tube to the fluid block.

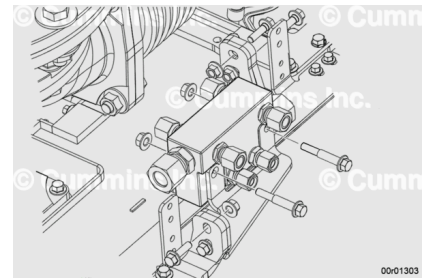


Remove and discard the o-rings on the male unions.

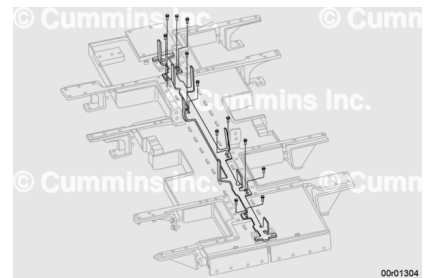


Note : To remove the spine bracket the coolant lines must be removed. Refer to Procedure 010-041 in Section 10 (/qs3/pubsys2/xml/en/procedures/56/56-010-041.html) to remove the coolant lines.

Remove the fluid manifold from the spine bracket.



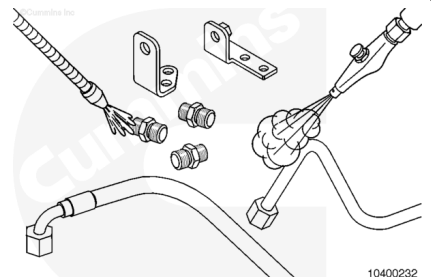
Remove the capscrews from the spine bracket.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

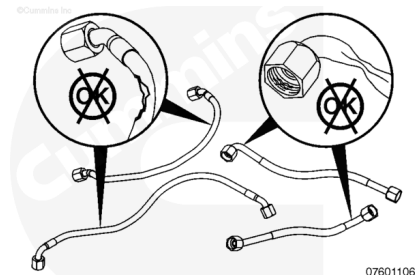
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

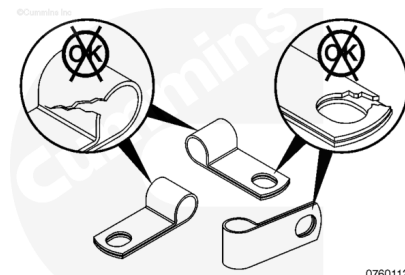
Wear appropriate eye and face protection when using compressed air. Flying debris or dirt can cause personal injury.

Clean the turbocharger oil supply lines and components with solvent and dry with compressed air.

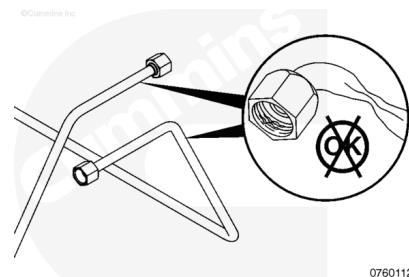
Inspect the turbocharger oil supply hoses for restrictions, cracks, or other damage. Replace if necessary.



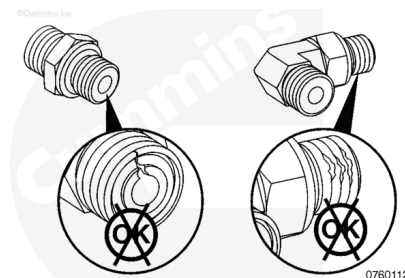
Inspect the p-clips. Replace if necessary.



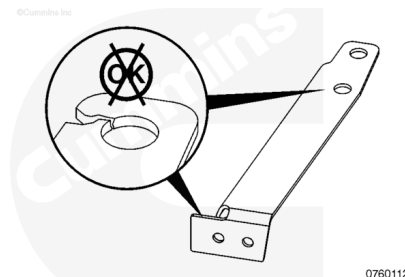
Inspect the turbocharger oil supply tubes for damage, kinks, or restrictions. Replace if necessary.



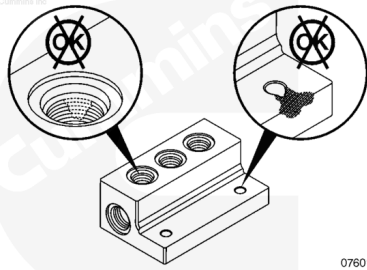
Inspect the male unions for stripped threads, cracks, or other damage. Replace if damaged.



Inspect the oil supply line braces for corrosion, cracks, or other damage. Replace if damaged.



Inspect the oil supply manifold for stripped threads, cracks, or other damage. Replace if necessary.



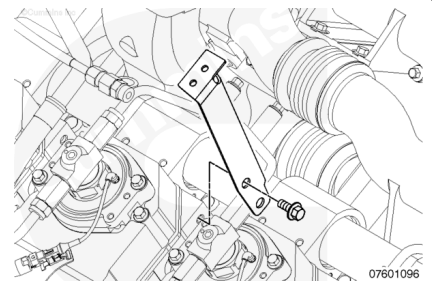
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Install

QSK45 and QSK60 Single Stage

⚠ CAUTION ⚠

When installing or reinstalling flexible hoses and tubing be sure that all the original factory installed clips are inspected and fitted. Reinstall so that the routing of fluid plumbing follows the original factory design and ensure the routing provides adequate clearances to prevent contact with any other engine components or high temperature engine areas. Inspect and reinstall any heat shielding to original factory configuration. Not providing adequate clearances between fluid plumbing and other engine components can lead to possible component damage, leakage and potentially hazardous conditions.



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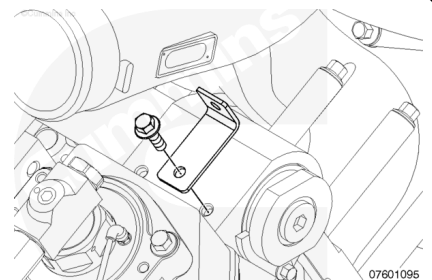
With Mechanically Actuated Injector

- Attach the braces to the engine block with the capscrews.

Torque Value: 80 n•m [60 ft-lb]

Attach the clips to the engine using the capscrews.

Torque Value: 45 n•m [33 ft-lb]



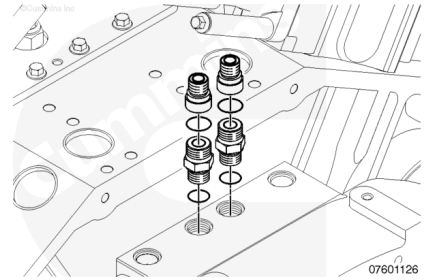
07601095

Install new o-rings on the male unions and install them into the turbochargers.

Torque Value: 60 n•m [44 ft-lb]

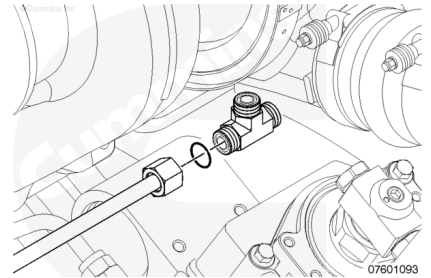
Install new o-rings on the male unions and install them into the flywheel housing.

Torque Value: 90 n•m [66 ft-lb]



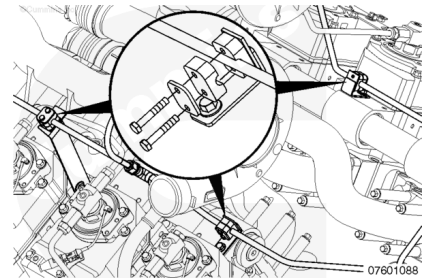
Install the T-fitting.

Torque Value: 45 n•m [33 ft-lb]



Install the oil supply line clamps.

Torque Value: 45 n•m [33 ft-lb]

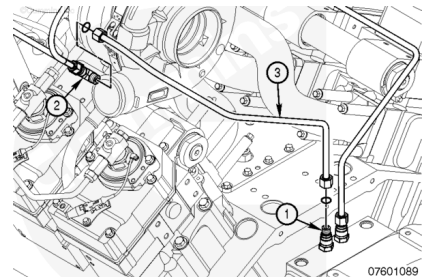


Attach the oil supply tube to the reducer (1).

Torque Value: 45 n•m [33 ft-lb]

Install a new o-ring on the T-fitting (2) and attach the other end of the oil supply tube (3) to the T-fitting.

Torque Value: 45 n•m [33 ft-lb]

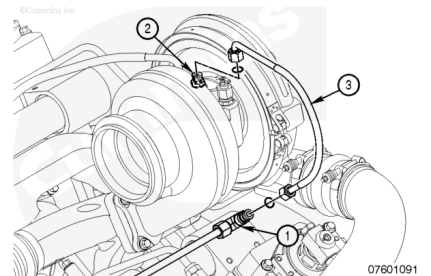


Attach the oil line (3) to the union (2) on the turbocharger.

Torque Value: 45 n•m [33 ft-lb]

Install a new o-ring and attach the oil supply hose to the T-fitting (1) union.

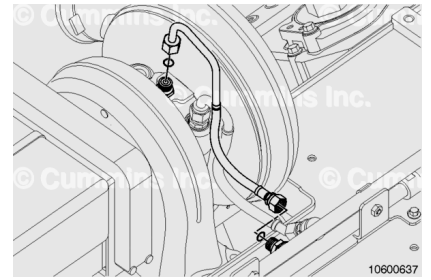
Torque Value: 45 n•m [33 ft-lb]



With Electronically Actuated Injector

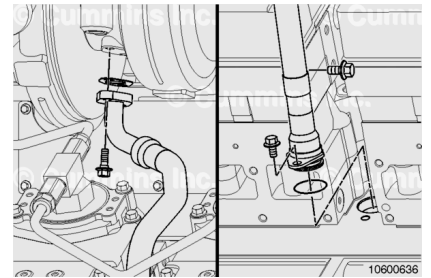
- Install new o-rings on both ends of the turbocharger oil supply tube.
- Install the turbocharger oil supply tube to the turbocharger and turbocharger oil manifold.

Torque Value: 45 n•m [33 ft-lb]



Secure the turbocharger oil supply tube with clamp.

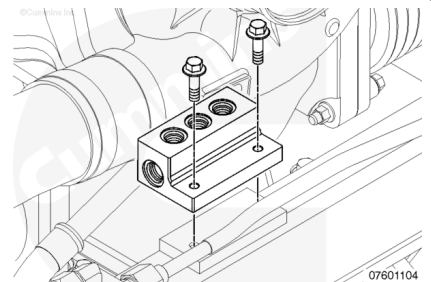
Torque Value: 45 n•m [33 ft-lb]



QSK60 Two Stage

⚠ CAUTION ⚠

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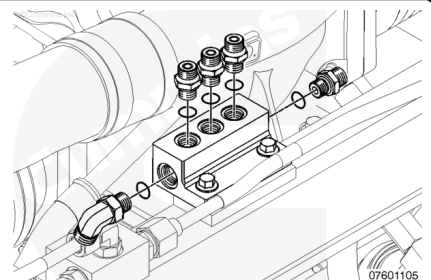


Install the oil supply manifold.

Torque Value: 45 n•m [33 ft-lb]

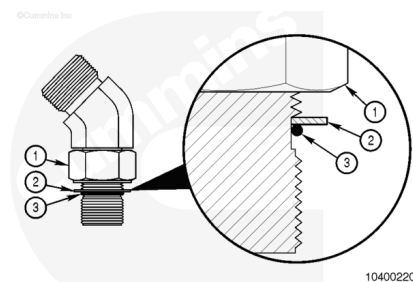
Assemble the oil supply manifold by installing new o-rings on the male unions and installing them into the supply manifold.

Torque Value: 55 n•m [41 ft-lb]

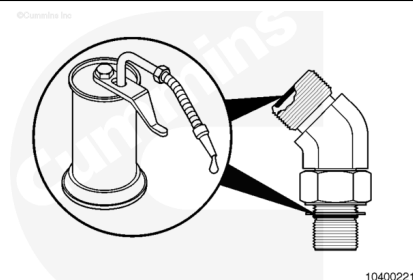


Install the male union elbow into the oil supply manifold by first positioning the locknut.

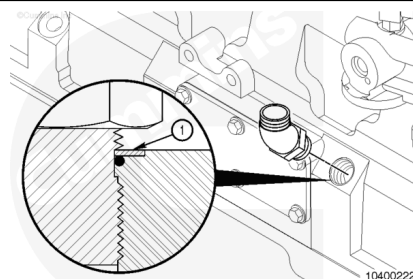
1. Locknut
2. Washer
3. O-ring.



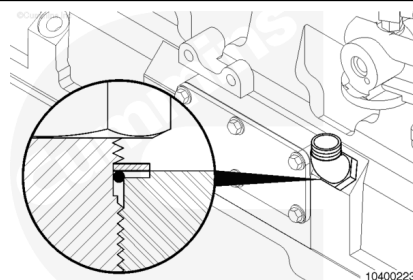
Install new o-rings on the union elbow and apply a light coat of clean engine oil to each o-ring.



Install the elbow union in the oil supply manifold until the backup washer (1) contacts the port face.



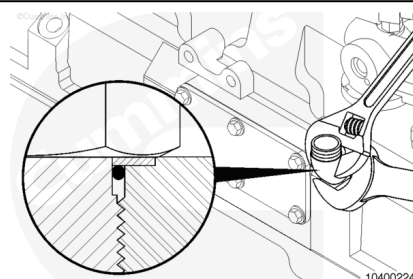
Position the elbow union to align the connection to the turbocharger oil supply line. Do **not** back off more than one full turn.



Inspect to verify the o-ring is **not** pinched and backup washer seats flat on the face of the port.

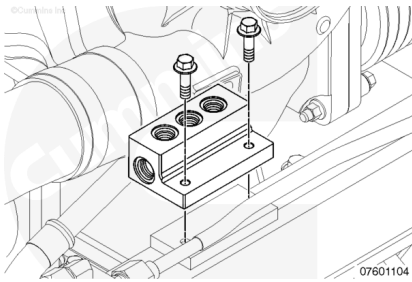
Use two wrenches to hold the male elbow in the desired position and tighten the locknut.

Torque Value: 80 n•m [60 ft-lb]



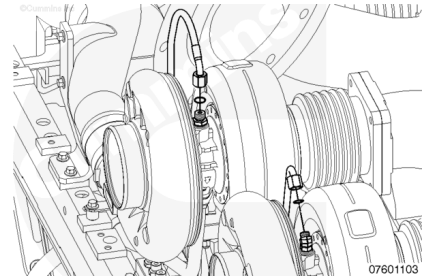
Install the assembled oil supply manifold to the engine block with two capscrews.

Torque Value: 45 n•m [33 ft-lb]



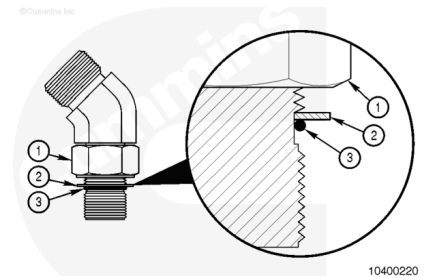
Install new o-rings on the male unions and install them into the turbochargers.

Torque Value: 45 n•m [33 ft-lb]

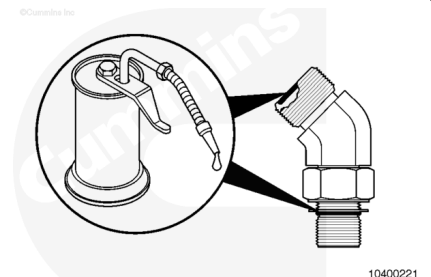


Install the male union elbow into the oil supply manifold by first positioning the locknut.

1. Locknut
2. Washer
3. O-ring.

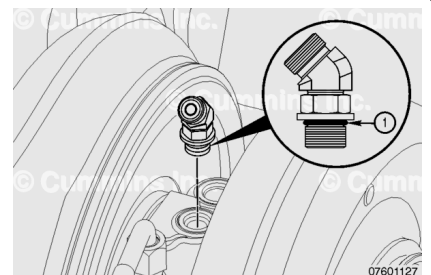


Install new o-rings on the union elbow and apply a light coat of clean engine oil to each o-ring.



Install the elbow union in the turbocharger until the backup washer (1) contacts the port face.

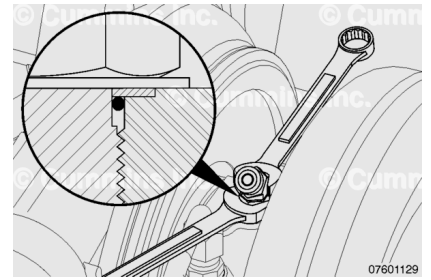
Position the elbow union to align the connection to the turbocharger oil supply line. Do **not** back off more than one full turn.



Inspect to verify the o-ring is **not** pinched and backup washer seats flat on the face of the port.

Use two wrenches to hold the male elbow in the desired position and tighten the locknut.

Torque Value: 45 n•m [33 ft-lb]

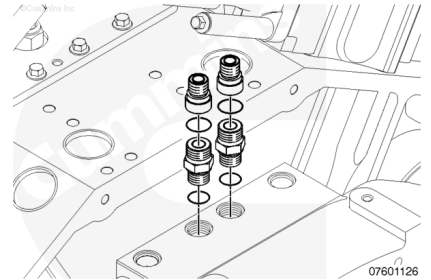


Install new o-rings on the male unions and install them into the flywheel housing.

Torque Value: 90 n•m [66 ft-lb]

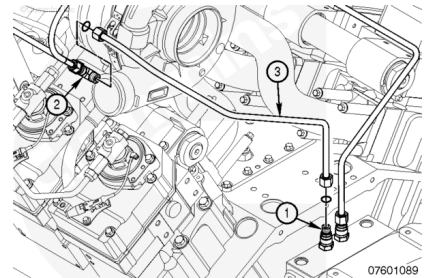
Install a new o-ring on the reducer and install it on the male union.

Torque Value: 60 n•m [44 ft-lb]



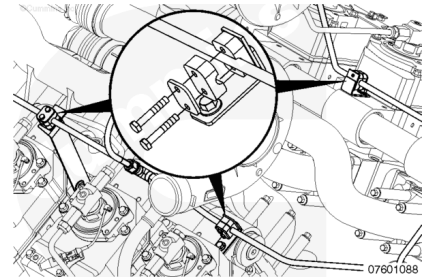
Attach the oil supply line to the reducer.

Torque Value: 45 n•m [33 ft-lb]



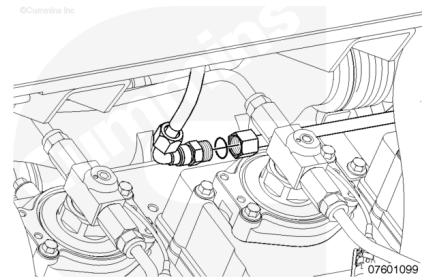
Install the oil supply line clips.

Torque Value: 45 n•m [33 ft-lb]



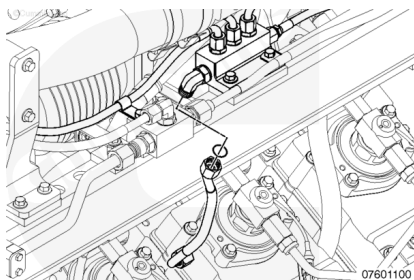
Install a new o-ring and attach the union elbow to the oil supply line.

Torque Value: 45 n•m [33 ft-lb]



Install a new o-ring onto each end of the oil supply hose and install onto the elbow and oil supply manifold.

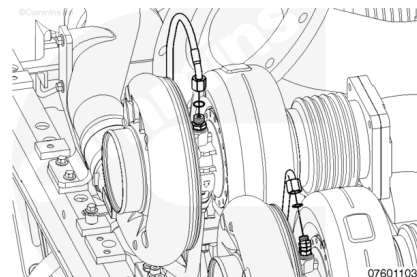
Torque Value: 45 n•m [33 ft-lb]



First Stage Turbochargers

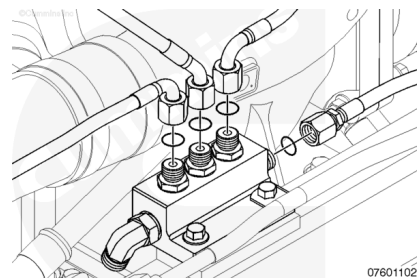
Install a new o-ring and attach the oil supply line to the union on the turbocharger.

Torque Value: 45 n•m [33 ft-lb]



Install a new o-ring on the union fitting on the oil supply manifold and attach the oil supply hose.

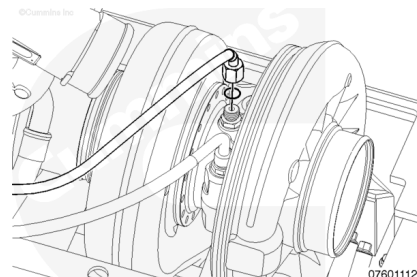
Torque Value: 45 n•m [33 ft-lb]



Second Stage Turbochargers

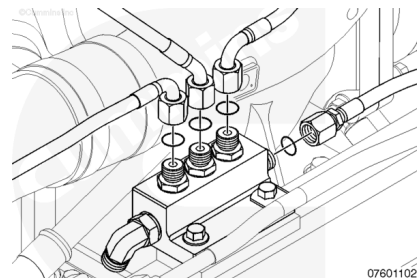
On the right bank rear turbocharger, install a new o-ring on the turbocharger union and attach the oil supply hose.

Torque Value: 45 n•m [33 ft-lb]



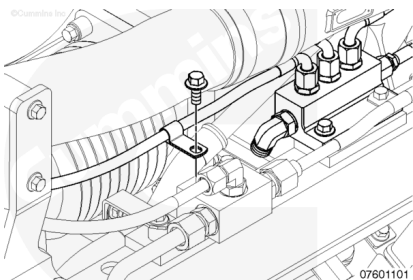
Install a new o-ring on the oil supply manifold and attach the oil supply hose.

Torque Value: 45 n•m [33 ft-lb]



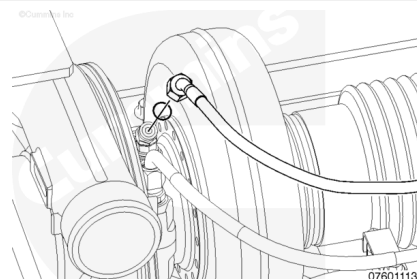
Attach the oil supply line p-clip.

Torque Value: 45 n•m [33 ft-lb]



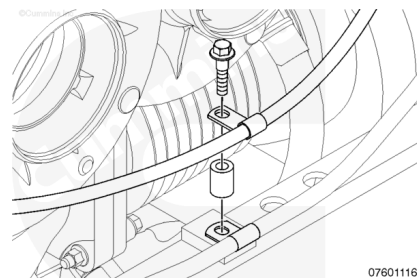
On the right bank front turbocharger, install a new o-ring on the turbocharger union and attach the oil supply hose.

Torque Value: 45 n•m [33 ft-lb]



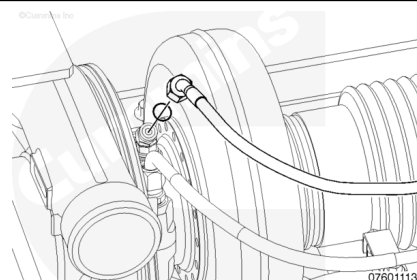
Attach the oil supply line p-clip.

Torque Value: 45 n•m [33 ft-lb]



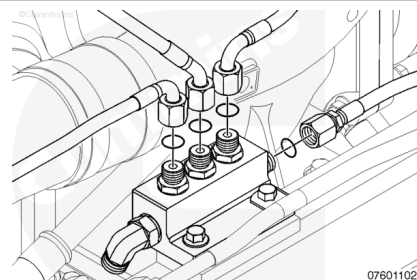
On the left bank rear turbocharger, install a new o-ring on the turbocharger union and attach the oil supply hose.

Torque Value: 45 n•m [33 ft-lb]



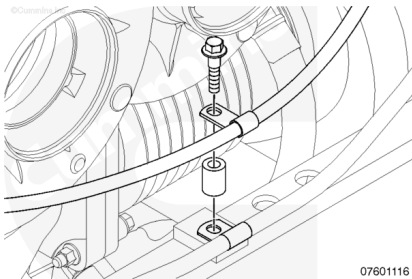
Install a new o-ring on the oil supply manifold and attach the oil supply hose.

Torque Value: 45 n•m [33 ft-lb]



Attach the oil supply line p-clip.

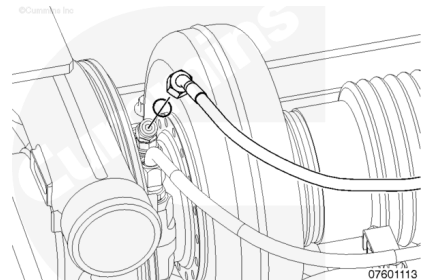
Torque Value: 45 n•m [33 ft-lb]



07601116

On the left bank front turbocharger, install a new o-ring on the turbocharger union and attach the oil supply hose.

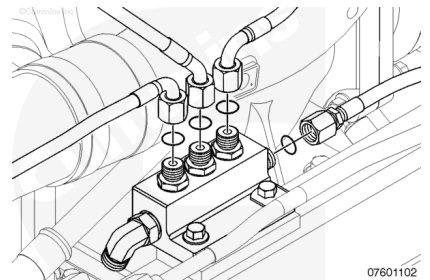
Torque Value: 45 n•m [33 ft-lb]



07601113

Install a new o-ring on the oil supply manifold and attach the oil supply hose.

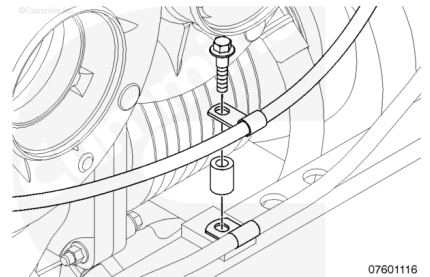
Torque Value: 45 n•m [33 ft-lb]



07601102

Attach the oil supply line p-clip.

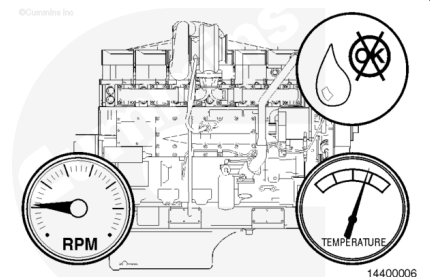
Torque Value: 45 n•m [33 ft-lb]



07601116

⚠ CAUTION ⚠

Confirm fluid tubing routing provides adequate clearances to prevent contact with any other engine components or high temperature engine areas. Confirm reinstallation of any heat shielding to original factory configuration. Not providing adequate clearances between fluid plumbing and other engine components can lead to possible component damage, leakage and potentially hazardous conditions.



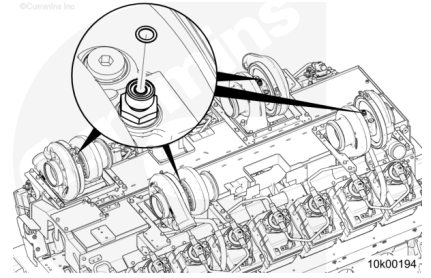
14400006

Operate the engine and check for leaks.

QSK45 and QSK60 Marine Single Stage With Flexible Lines

⚠ CAUTION ⚠

When installing or reinstalling flexible hoses and tubing be sure that all the original factory installed clips are inspected and fitted. Reinstall so that the routing of fluid plumbing follows the original factory design and ensure the routing provides adequate clearances to prevent contact with any other engine components or high temperature engine areas. Inspect and reinstall any heat shielding to original factory configuration. Not providing adequate clearances between fluid plumbing and other engine components can lead to possible component damage, leakage and potentially hazardous conditions.



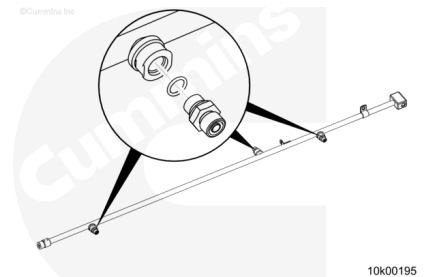
Install new o-rings on the male unions and install them into the turbochargers.

Tighten each end of the flexible hose.

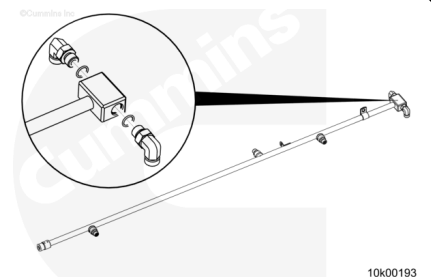
Torque Value: 45 n•m [33 ft-lb]

Install three male unions to the oil transfer tube.

Torque Value: 45 n•m [33 ft-lb]

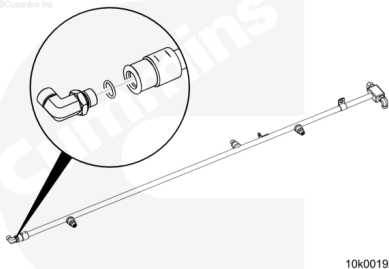


Fit two elbow male unions to the rear ports of the oil transfer tube hand tight.



Fit one elbow male union to the front port of the oil transfer tube hand tight.

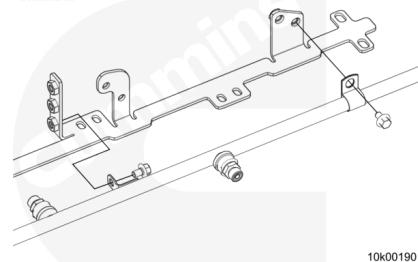
Torque Value: 45 n•m [33 ft-lb]



10k00192

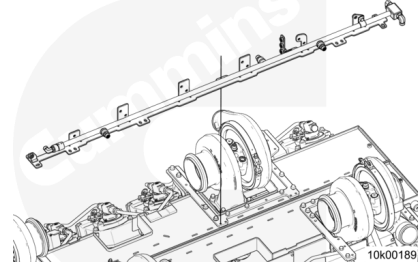
Install the oil transfer tube to the transfer tube bracket. Use the P-clip cap screw and L-bracket cap screw.

Torque Value: 45 n•m [33 ft-lb]



10k00190

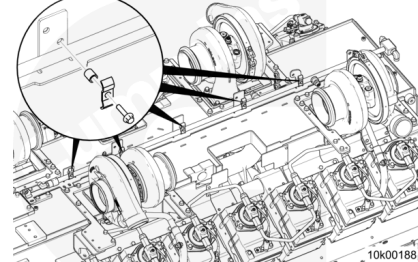
Install the oil transfer tube and bracket assembly onto the engine.



10k00189

Secure oil and water transfer tubes with five capscrews through tube clamps, spacers into tube bracket.

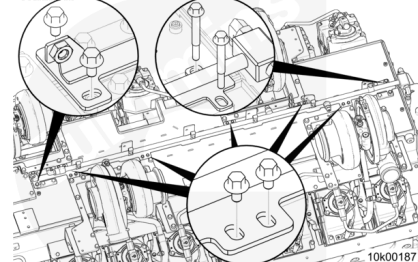
Torque Value: 22 n•m [195 in-lb]



10k00188

Secure oil and water transfer tube bracket into place. Install capscrews.

Torque Value: 45 n•m [33 ft-lb]

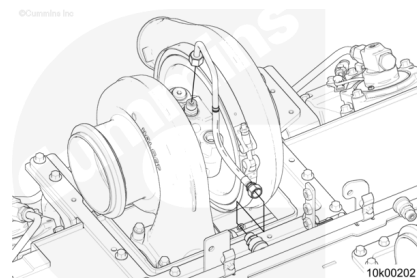


10k00187

Right bank rear and left bank front. Make sure o-ring is in straight male union prior to connecting flexi hose. Connect oil transfer flexible hose to the outboard male union. Connect

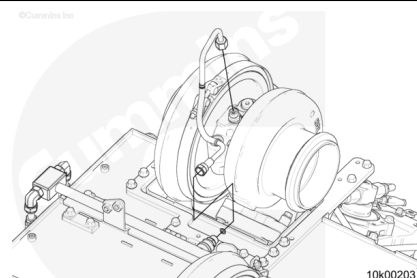
bottom first and then top connection. Make sure lines are routed as shown in the diagram and lines are routed away from turbocharger turbine housing.

Torque Value: 33 n•m [24 ft-lb]



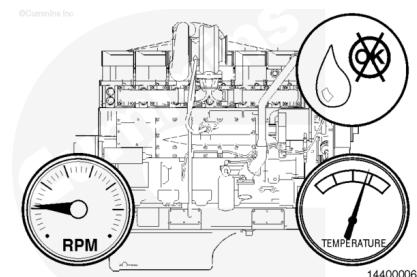
Right bank front and left bank rear. Make sure o-ring is in straight male union prior to connecting flexible hose. Connect oil transfer flexi hose to the outboard male union. Connect bottom first and then top connection. Make sure lines are routed as shown in the diagram and lines are routed away from turbocharger turbine housing.

Torque Value: 33 n•m [24 ft-lb]



⚠ CAUTION ⚠

Confirm fluid tubing routing provides adequate clearances to prevent contact with any other engine components or high temperature engine areas. Confirm reinstallation of any heat shielding to original factory configuration. Not providing adequate clearances between fluid plumbing and other engine components can lead to possible component damage, leakage and potentially hazardous conditions.

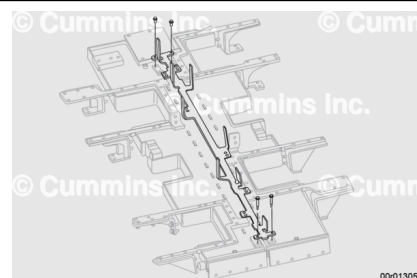


Operate the engine and check for leaks.

QSK45 and QSK60 Marine Single Stage With Rigid Lines

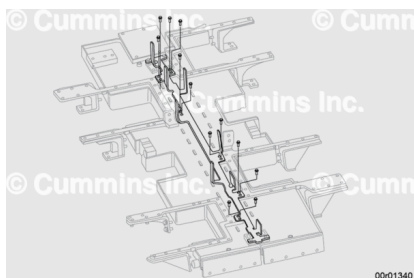
Install four capscrews to the front and rear of the support bracket.

Torque Value: 45 n•m [33 ft-lb]

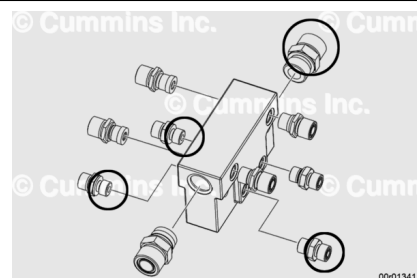


Install 12 capscrews to the center of the support bracket.

Torque Value: 22 n•m [195 in-lb]



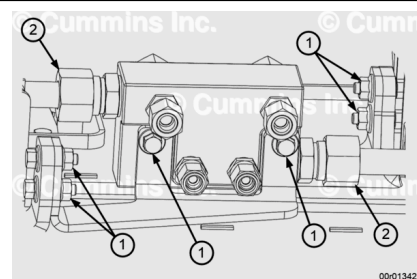
Install new o-rings to fluid manifold connections and turbocharger connections.



Install fluid manifold and lubricating oil supply tube. Tighten all nuts.

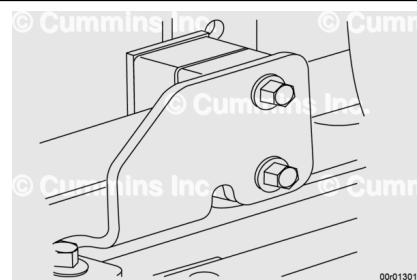
Torque Value:

1. 115 n•m [85 ft-lb]
2. 45 n•m [33 ft-lb]



Install the two rail tube clamps.

Torque Value: 9 n•m [80 in-lb]



Attach T fitting to end of lubricating oil supply tube.

Torque Value:

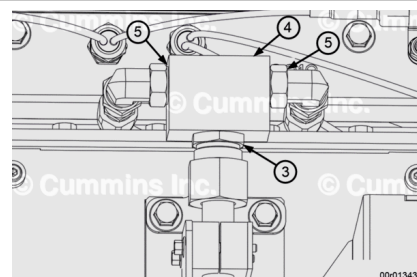
(3) 115 n•m [85 ft-lb]

Torque Value:

(4) 97 n•m [72 ft-lb]

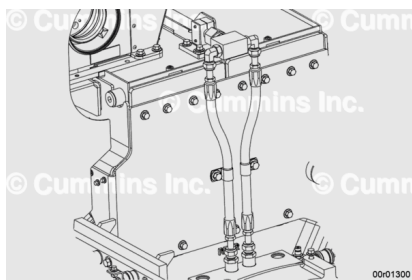
Torque Value:

(5) 45 n•m [33 ft-lb]



Attach lubricating oil supply hoses to the elbows.

Torque Value: 61 n•m [45 ft-lb]



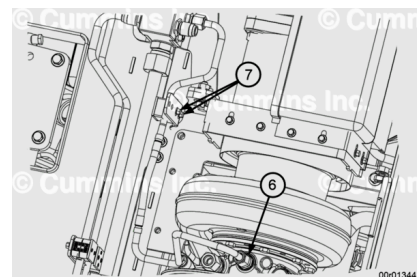
Install turbocharger lubricating oil supply tube and clamping block.

Torque Value:

(6) 45 n•m [33 ft-lb]

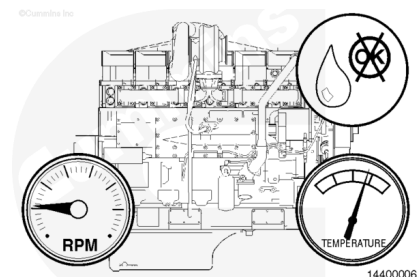
Torque Value:

(7) 10 n•m [89 in-lb]



⚠ CAUTION ⚠

Confirm fluid tubing routing provides adequate clearances to prevent contact with any other engine components or high temperature engine areas. Confirm reinstallation of any heat shielding to original factory configuration. Not providing adequate clearances between fluid plumbing and other engine components can lead to possible component damage, leakage and potentially hazardous conditions.



Operate the engine and check for leaks.